

Dante M. Benedetti

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Education

California Polytechnic State University, San Luis Obispo

September 2022 - Present

- **M.S. Electrical Engineering** - Expected May 2027
- **B.S. Electrical Engineering** - June 2026 | GPA 3.89 | **Dean's List** (12 Quarters)

Skills

Power Electronics: Four-Switch Buck-Boost & MPPT Converters, Peak-Current-Mode Control, Slope Compensation, Gate Drive, Loop Compensation & Bode Analysis, Multilayer PCB Layout, Grounding & Return-Path Design

Software/Tools: KiCad (Schematic Capture & PCB Layout), SIMetrix/SIMPLIS, LTspice, Fusion 360 (PCB Design), Python, ROS 2, STM32CubeIDE, Vivado, AutoCAD, Git

Controls & Autonomy: PID, State-Space (Pole Placement, Luenberger Observer), Extended Kalman Filter, Sensor Fusion, Theta* Path Planning, Pure Pursuit, Stanley Control

Perception: YOLO Object Detection, OpenCV, RGB-D Perception, LiDAR Mapping

Hardware: STM32/Nucleo, Jetson Orin Nano, IMU, Magnetometer, Optical Flow, Time-of-Flight, Robotic Manipulation, Oscilloscope, Function Generator, Digital Power Supply, Solder

Technical Projects

PCB Power System for Satellite MPPT: *M.S. Project, Cal Poly, San Luis Obispo*

September 2026 - Present

- **Cradle-to-grave ownership** of a 30 W four-switch buck-boost converter regulating a 6-24 V spacecraft solar array to a 7.4 V battery bus: design, simulation, layout, fabrication, and bench test
- Closed the MPPT loop entirely in analog hardware (300 kHz peak-current-mode, slope compensation, PI outer loop); verified in SIMetrix/SIMPLIS - bus holds within 96 mV as the array sweeps 10.3 V to 5.0 V
- Captured an 8-sheet schematic and routed a 132-component, 4-layer board in KiCad with split switching and analog grounds tied at a single star point; fabrication and bench verification underway

Autonomous Vision-Based Litter Collection Rover: *Senior Project, Cal Poly, San Luis Obispo*

September 2025 - June 2026

- Designed and built a three-person autonomous rover capable of detecting, navigating to, and collecting ground-level litter
- Integrated an NVIDIA Jetson Orin Nano running ROS 2 with an STM32 real-time control platform driving mecanum-wheel mobility and a two-degree-of-freedom collection mechanism
- Implemented YOLO-based object detection, LiDAR mapping, EKF state estimation, Theta* path planning, and pure-pursuit path tracking

Autonomous Quadrotor Flight Controller: *Independent Summer Project*

June - August 2026

- Designed and built a custom 232 mm quadrotor platform using 5-inch propellers and a sensor-fusion-based flight-control architecture
- Developed PID attitude and altitude control using IMU, magnetometer, optical-flow, and time-of-flight sensing, with EKF state estimation for autonomous position control

Vision-Based Robotic Manipulator: *Cal Poly, San Luis Obispo*

September - December 2025

- Implemented image-processing and robotic-control algorithms for a four-joint robotic arm using object-oriented Python
- Developed trajectory-generation methods and evaluated forward and inverse kinematic models for autonomous manipulation tasks

WERC Environmental Design Contest: *Cal Poly, San Luis Obispo*

September 2024 - April 2025

- Researched, designed, and built multiple drafts of a scalable, low-cost green-hydrogen energy-storage system to optimize electrical grid stability; presented the final design to a board of judges at NMSU
- Placed First for Task 2 Overall Design; published "Sustainable Grid Stabilization Via Green Hydrogen Technologies" through IEEE Xplore

Relevant Courses

Power Electronics & Energy: Power Electronics (Buck, Boost & Buck-Boost Converter Design), Solar Energy Conversion, Analog/Digital Electronics, Semiconductor Device Electronics

Controls & Autonomous Systems: Control Systems Theory, Classical/Digital Control Systems, Planning & Control for Autonomous Vehicles

Robotics & Vision: Computer Vision, Vision-Based Robotic Manipulation, Autonomous Vehicles Laboratory

Embedded & Digital Systems: Microcontroller-Based System Design, Computer Digital Design, Computer Design & Assembly Language, Data Structures

Professional Experience

Project Engineer, Pribuss Engineering INC. *South San Francisco*

June 2025 - September 2025

- Designed multi-level, San Francisco City approved, fire sprinkler systems using AutoCad's AutoSprink
- Managed and coordinated on-site inspections, material delivery, and city-based hydraulic calculations